

Honeywell

TPN Backup and Restore

User's Guide

TP23W-250

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About This Document

This document describes the TPN Backup and Restore tool and provides the procedures for performing TPN backup and restore operations.

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References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title
Engineer's Reference Manual, SW09-605

Contacts

World Wide Web

The following Honeywell web sites may be of interest to Process Solution customers.

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION Identifies information that requires special consideration.
	CAUTION Indicates a situation which, if not avoided, may result in work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	TIP Identifies advice or hints for the user, often in terms of performing a task.

	REFERENCE –EXTERNAL Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

Symbol Definitions

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Contents

1. Introduction

1.1 About This Section

This section introduces the software tools available to you for performing backup and restore operations.

1.2 Available Backup and Restore Tools

The following software tools are available to you for protecting your system from unexpected events:

- TPN Backup and Restore for a TPN System
- Experion Backup and Restore (EBR) for a TPS System and an Experion System

1. Introduction

1.3. TPN Backup and Restore Tool

1.3 TPN Backup and Restore Tool

You can use the TPN Backup and Restore tool to backup the TPN History Module (HM) image from one or more HM hard drives to a Global User Station (GUS) or an Application Processing Platform (APP), as shown in Figure 1-1. The tool can also be used to restore one or more HM hard drives from a saved backup.

To use this tool, the File Transfer function must be active in the TPS system.

The backup files can also be saved as .LCN files so that they can be accessed as .LCN files by a TPS node.

Specifically, you can use this tool for:

- Configuring and modifying a backup configuration
- Deleting a backup configuration
- Scheduling daily, weekly, and monthly backups
- Modifying a backup schedule
- Unscheduling a backup
- Demanding a backup
- Monitoring the progress of a backup
- Requesting that data-point checkpoints be part of the backup
- Creating .LCN files from the backup files
- Restoring the image, volumes, directories, and files to an HM

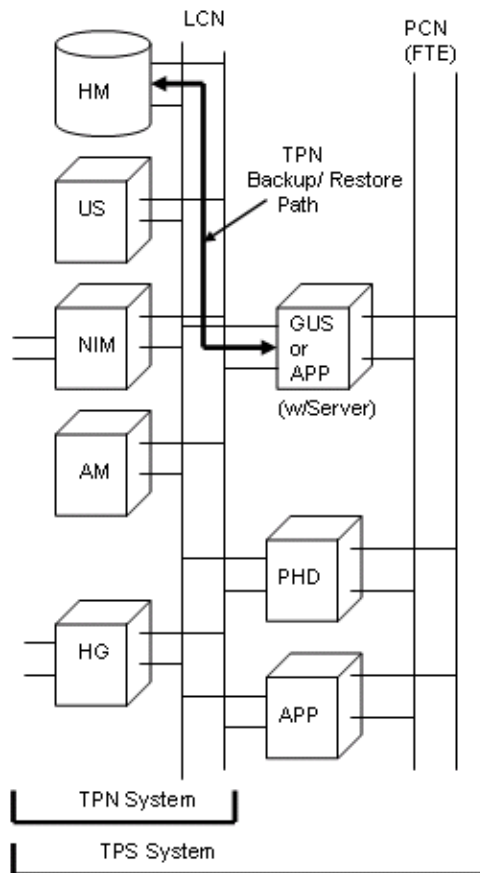


Figure 1-1 TPN Backup and Restore Path

1. Introduction

1.4. Experion Backup and Restore Tool

1.4 Experion Backup and Restore Tool

Figure 1-2 is a topological diagram that shows the EBR backup and restore paths in the TPS System, and in the Experion System. Figure 1-2 also shows the backup and restore path for the TPN Backup and Restore tool to provide you with a comprehensive overview of the complete backup and restore paths across the systems.

For more information about EBR, refer to the *Experion Backup and Restore User's Guide*, EP-DCX424.

Using EBR for the TPS System

If you have a TPS System, you can use EBR to backup the images of the Global User Station (GUS), Domain Controller, and the Process Historian Database (PHD) nodes to a node that you designated as the TPS EBR file server as shown in Figure 1-2. After the images have been copied to the TPS EBR file server, you can then copy the images to a CD, DVD, or magnetic tape for storage at a secure off-site location.

You can also restore the image to each node from the TPS EBR File Server using EBR as shown in Figure 1-2.

Using EBR for the Experion System

Using EBR, you can perform a complete backup and restore of the nodes and databases of your Experion System.

In the sample Experion System shown on the right side of Figure 1-2, the images of the ACE Supervisory controller, redundant server, Station, ESVT, and ES-T can be backed-up to an EBR File Server. You can also copy the images from the EBR File Server to a DVD, CD, or magnetic tape for storage in a secure off-site location.

You can also restore the image to each node from the Experion EBR File Server using EBR, when required.

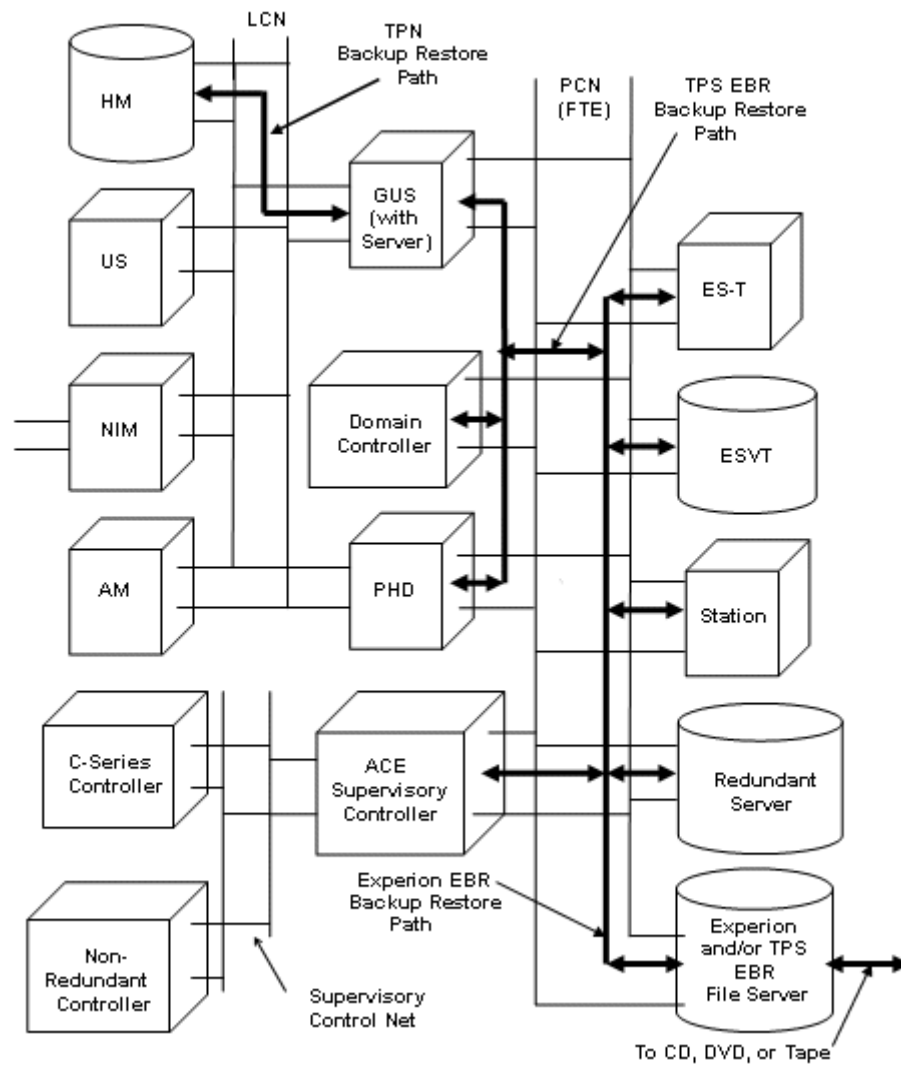


Figure 1-2 TPN, TPS, and Experion Backup and Restore Paths

1. Introduction

1.4. Experion Backup and Restore Tool

2. Developing Backup and Recovery Strategies

2.1 About this Section

This section provides recommendations to consider when you develop comprehensive backup and recovery strategies before you begin to use backup and restore tools. It would be beneficial to have system administrators and users participate in developing the strategies.

After you develop your backup and recovery strategies, you can use the backup and restore tools described in this document to implement your strategies.

2.2 Developing a Backup Strategy

All the recommendations below may not apply to the available backup tools, but this will give you an overview of the type of information that you will need to have before you begin implementing your backup strategy.

Determining nodes to be used

You must know the nodes and the node types in your system so that you can specify the following information:

- What system nodes are to be backed up?
- What drives are to be backed up in each node?
- What node will be used as the server that retains all the backup images?

Scheduling backups

Scheduling the frequency of backups normally depends on the following factors:

- Rate of data change (configured data)
- Database availability
- Value of the data to the company

Table 2-1 is a sample worksheet that you may want to consider developing as a spreadsheet to make your task of scheduling backups easier by having all the information in one place.

2. Developing Backup and Recovery Strategies

2.2. Developing a Backup Strategy

Table 2-1 Sample backup schedule worksheet

Node Type	Schedule				Comments
	Full Backup	Incremental Backup	Time between Incremental Backups	Consolidate Incremental Backups	

Checkpointing data repositories

When you are scheduling backups, ensure that you coordinate your backup scheduling with the checkpoints that must be performed for TPN nodes. Ensure that manual checkpointing of TPN nodes is performed before the backup jobs begin.

If you are using the TPN Backup and Restore tool, checkpointing can be scheduled so that it is performed automatically as the first part of the backup job.

In addition, if you are using the TPN Backup and Restore tool and another tool such as Experion Backup and Restore, allow sufficient time for the TPS/TPN backup to be finished before you schedule the Experion Backup and Restore tool to begin the backup job for the Experion nodes.

Staggering start times

When scheduling backup jobs for multiple nodes, it is strongly recommended that the start times be staggered so that normally only one node is being backed up at a time.

Other backup considerations

To ensure that your backups run successfully, you should observe the backup considerations described below.

Rebuilding or restoring data

Do not rebuild or restore a database at the same time that a backup is being performed. Database corruption can occur when the node is low on hard-disk space and you are doing a database rebuild when a backup is in progress.

Running disk I/O intensive applications

Running disk I/O intensive applications at the same time that a backup is being performed can have a significant performance impact on the time required to complete the backup and/or to complete the other application. It is best to avoid using these applications at the same time.

Running disk defragmentation software

Do not run disk defragmentation software when a backup is in progress. It will cause the backup image to be invalid and you will not be able to restore the drive. Also, do not run any partitioning software (such as Norton PartitionMagic™) at the same time as a backup is being performed.

Running general applications

Some general applications such as ScanDisk must be turned off when a backup is in progress.

You should also avoid Honeywell-specific actions such as server synchronization, archiving/restoring to or from a tape, and file transfer to/from the LCN.

Off-site storage

It is recommended that you transfer your backup images onto CD or DVD ROM, or tape on a regularly-scheduled basis. You should store the media off-site in a secure temperature-controlled environment for long-term storage. You should also define the backup retention period for the media.

2.3 Developing a Recovery Strategy

You also need to develop a system recovery strategy that includes the following considerations:

- What are the possible disaster scenarios that can affect your process and system?
- Depending on the disaster scenarios, what are the possible recovery procedures?
- What is the recovery procedure for each scenario?
- Who performs the recovery tasks?
- Should you develop a schedule for the recovery?
- If your plant is divided into operational areas, is there a hierarchy of operational areas to be restored/restarted?
- What is the order in which the system nodes within a cluster or operating area are to be restarted/restored?
- Where are your backup images to be stored? Are they available from a nearby secure, temperature-controlled off-site location?

The system recovery plan should be documented and available to all possible participants in the recovery scenarios.

3. Installing TPN Backup and Restore

3.1 About this section

This section describes the software and hardware pre-requisites for the TPN Backup and Restore tool, and provides you with installation information.

3.2 Installation Pre-Requisites

Software and Hardware

To use the TPN Backup and Restore tool, your system must have the following minimum software and hardware:

- GUS Release 350
- APP Release 240
- Windows 2000 platform with LCNP4
- Honeywell File Transfer tool (June 2004 version, or later)

Note that the File Transfer tool is a separately licensed product

3.3 Installing the TPN Backup and Restore Tool

Installable Packages

You can install the TPN Backup and Restore tool as a package when you install one of the following Honeywell CD-ROMs:

- Experion CD-ROM
- TPS System CD-ROM
- App Node CD-ROM

3. Installing TPN Backup and Restore

3.4. Access Levels



ATTENTION

The File Transfer tool is also an installable package on the same CD-ROMs. You must install the File Transfer tool in the same GUS or APP node together with the TPN Backup and Restore tool. **The TPN Backup and Restore tool will not work without having File Transfer installed in the same node.**

Note that you will not be able to restore file descriptors using the TPN Backup and Restore tool because File Transfer does not copy file descriptors.

3.4 Access Levels

The interactive interface, including the ability to configure, schedule, and demand backups and the ability to restore to the History Module is available only if you are a member of the Administrators and TPS Engineers groups.



ATTENTION

Each of the accounts used for TPN Backup and Restore operations must have full control access to the folders configured as the destinations for log files, backup files, and .LCN files.



CAUTION

When multiple instances of the TPN Backup and Restore user interface are run at the same time, caution needs to be exercised to avoid making inconsistent configuration changes. The primary use for running multiple copies is to allow you to view/alter configurations while the other is running a demand backup or restore.

3.5 TPN Backup and Restore Icon

When a scheduled or demanded backup is executing, the icon shown in Figure 3-1 appears in the system tray.



Figure 3-1 Backup-in-Progress Icon

Clicking on the icon opens a status dialog. The status dialog identifies the file currently being processed, the progress of the process, and errors detected. You can use this dialog to monitor demand backups.

This system tray icon can also be used to stop a backup in progress. An aborted backup during the backup process retains the files that were already copied, enters a log entry indicating that the backup was aborted by the user and does not create a .LCN file from the partial backup.

For an aborted backup during .LCN file creation (when the backup files have already been completed), the .LCN file contains only the files that have been already copied. The log file will indicate that the .LCN file creation was aborted by the user.

The icon changes color depending on the status of the backup as follows:

- Yellow – backup has started
- Green – a successful connection was made to the File Transfer tool.
- Red – an error has been detected

3.6 Accessing TPN Backup and Restore

Step	Action
1	Click the Start button.
2	Select Programs .
3	Select Honeywell TPS .
4	Select TPN Backup Restore . TPN Backup/Restore page appears as shown in Figure 3-2.

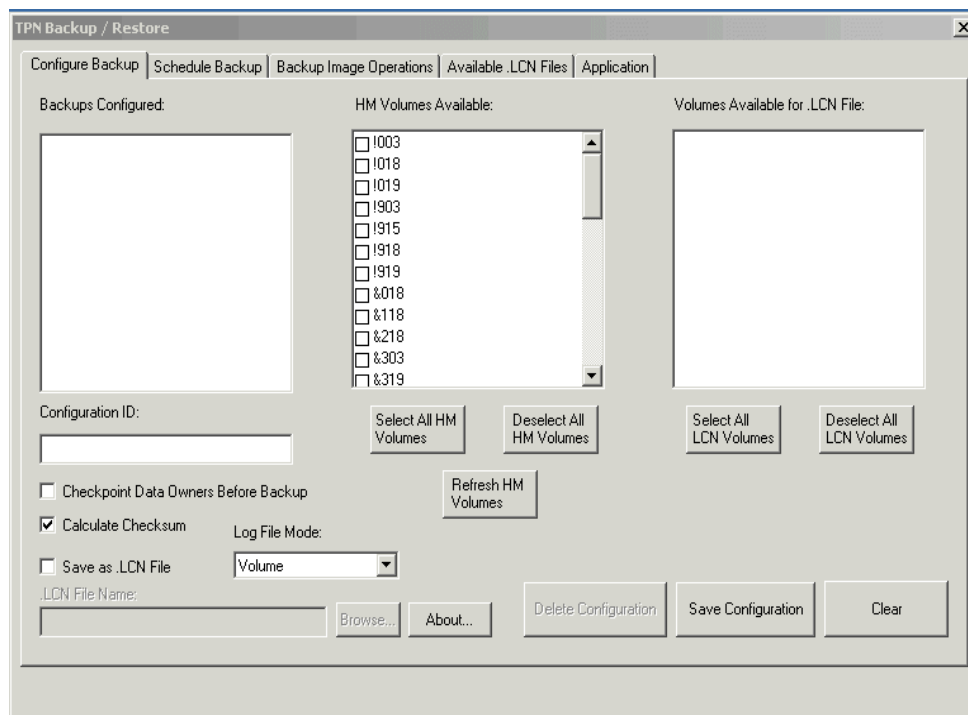


Figure 3-2 TPN Backup/Restore Page

3.7 Establishing Application Defaults

Establishing Default Pathnames and Logging Selection

You should establish default values for the following items before using the TPN Backup and Restore tool:

- Location where backup files are to be stored
- Location where configuration and log files are to be stored
- Location where .LCN files are to be stored

Step	Action
1	Select the Application tab, if it is not currently displayed. Application tab appears.
2	Type in the Pathname for Backups of the base directory that is to contain the saved backup files. Use the Browse button if you want to find a different directory. Pathname appears in text box.
3	Type in the Pathname for Configuration and Log Files of the base directory that is to contain the configuration and log files. Use the Browse button if you want to find a different directory. Pathname appears in text box.
4	Type in the Pathname for .LCN Files of the base directory that is to contain .LCN files. Use the Browse button if you want to find a different directory. Pathname appears in text box.
5	Click the Restore Log Enabled option if you want to have a continuous log of restore activities. A continuous restore log will be generated during restore activities. The View Restore Log and Clear Restore Log buttons are available if you want to view or clear the restore log.

3. Installing TPN Backup and Restore

3.7. Establishing Application Defaults

Step	Action
6	Click the Save button to save your pathname entries and the selected Restore Log Enabled option. Entries are saved.
7	If you want to continue with the next procedure, go to step 1 of the next procedure. If you want to exit the TPN Backup and Restore tool, select the Exit Application button. If you have not saved your entries, the entries will be lost.

4. Using the TPN Backup and Restore Tool

4.1 About this section

This section provides you with procedures for:

- Configuring a backup
- Modifying and deleting a configured backup
- Scheduling a backup
- Demanding a backup
- Monitoring backup progress
- Deleting a backup
- Viewing log information

4.2 Configuring a Backup

Creating a new backup configuration



ATTENTION

Selecting the **Clear** button on the **Configure Backup** tab cancels the current entries in this tab. Using the **Clear** button also returns the tab entries to the default settings.

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Configure Backup tab Configure Backup tab appears.
2	Type in the Configuration Id in the text box to identify this new backup. The identifier name for the backup should also be meaningful to other users. Examples of appropriate names are All Volumes , All but Personalities , and User Volumes Only . Name of backup appears in the text box.

4. Using the TPN Backup and Restore Tool

4.2. Configuring a Backup

3	Specify the History Module (HM) volumes to be included in the backup by clicking the Select All HM Volumes button. The names of all HM volumes appear in the list in the HM Volumes Available window. Each volume has been automatically selected for you. If you want to change the volume selection, go to step 4. If the volumes to be backed up are acceptable to you, go to step 5.
4	Deselect the HM volumes that you do not want in the backup file by clicking each box individually. An alternate method is to first select the Deselect All HM Volumes button and then click on only the volumes that you want. The volume names that are to be in the backup file that you are configuring are shown as selected in the HM Volumes Available window.
	ATTENTION If you have multiple HMs in your system being backed up using different backup configurations, selecting the Checkpoint Data Owners Before Backup option for one HM causes the checkpoint of all data owners to all HMs.
5	Select the Checkpoint Data Owners Before Backup option if you want all data owners such as NIMs and HGs on the TPN to be automatically checkpointed to the HM before the backup process begins. This ensures that you have the most current TPN checkpoints in the HM before the backup process begins. Do not select this option if checkpoints are not to be automatically created before the backup starts.
6	Select the Log File Mode from a drop-down menu. The Log File Mode specifies the logging mode (Volume, Directory, or File). The menu has the following selections: • Volume – log file will contain start time, end time, location of backup files, location of .LCN files (if the .LCN file option is selected), checkpoint pass/fail status for each checkpointed node including volumes, and errors. • Directory – log file will contain start time, end time, location of backup files, location of .LCN files (if the .LCN file option is selected), checkpoint pass/fail status for each checkpointed node including volumes and directories, file counts, and errors.

4. Using the TPN Backup and Restore Tool

4.2 Configuring a Backup

	• File - log file will contain start time, end time, location of backup files, location of .LCN files (if the .LCN file option is selected), and checkpoint pass/fail status for each checkpointed node including volumes, directories, files, and errors. The selected log file mode appears in window.
7	Select the Calculate Checksum option if you want have a checksum calculated for each file at the time of the backup, and a checksum calculated for each file at the time of the restore request. If the checksums are different, you are notified of the error. The Calculate Checksum option is selected as a default. If you do not want checksums to be calculated, deselect the option.
8	If you have selected the Save as LCN File option, you must also specify the name of the .LCN file that is to be created when the backup is saved. Type in the .LCN file name into the text box and/or use the Browse button. If the pathname (directory) portion of the .LCN file name is not specified, the default .LCN pathname specified on the Application tab will be used as the pathname.
9	Click Save as .LCN File option to convert the backup files to the .LCN file format. This is in addition to the normal backup files that will be in the PC directories. When selected, the Volumes Available For .LCN File list box is filled in with the available volume names. If the option is not selected, the Volumes Available For .LCN File list is empty. (This is the default option.)
10	If you have selected the Save as .LCN File option, you must specify the History Module (HM) volumes to be included in the backup. The same volumes that you have selected in the HM Volumes Available edit box also appear in the Volumes Available For .LCN File edit box. When configuring a new backup, all the volumes in the list are selected. Deselecting any volume in the HM Volumes Available list causes the volume to be removed from the Volumes Available For .LCN File list. When necessary, use the Select All LCN Volumes or Deselect ALL LCN Volumes buttons as required.
11	To save the configuration, click the SAVE CONFIGURATION button. Configuration ID name appears in the Backups Configured window and the backup configuration is saved in the location specified by the Pathname for Configuration and Log Files entry in the Application tab.

4. Using the TPN Backup and Restore Tool

4.2. Configuring a Backup

Modifying an existing backup configuration

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Configure Backup tab. Configure Backup tab appears.
2	Select an existing backup configuration ID from the Backups Configured list. Selected backup configuration ID is highlighted, and the elements of the configured backup appear.
3	To modify the HM volume configuration, select or deselect HM volume checkboxes shown in the HM Volumes Available list.
4	To modify the log file mode, select the Log File Mode drop-down menu and select Volume , Directory , or File . New selection appears in the window.
5	To modify the Checkpoint Data Owners Before Backup option, select or deselect the option.
6	To modify the Calculate Checksum option, select or deselect the option.
7	To modify the Save as .LCN File option, select or deselect the option.
8	To modify the .LCN File Name, type in the modified file name in the .LCN File Name text box. Revised .LCN file name appears.
9	To modify the volumes in the Volumes Available For .LCN File list, select or deselect the appropriate volume.
10	To save the backup configuration changes, select the Save Configuration button. Backup configuration changes have been saved.

Deleting an existing backup configuration

Step	Action
1	Select the Configure Backup tab of the TPN Backup / Restore page. The Configure Backup tab appears.
2	Select the name of the backup configuration that you want to delete from the Backups Configured list. Name of backup configuration is highlighted.
3	Select the Delete Configuration button. A pop-up message appears stating that: The command cannot be undone. Are you sure you want to Delete "Backup"? Click Yes . Backup configuration is deleted from the Backups Configured list.

4. Using the TPN Backup and Restore Tool

4.3. Scheduling Backups

4.3 Scheduling Backups



CAUTION

Anti-Virus disk scans and TPN backup tasks may interfere with each other. Avoid scheduling TPN backup tasks at times that may overlap with scheduled virus scans. You should also avoid concurrent operation of TPN backup and restore tasks with disk defragmentation utilities and TPS backups of the same node.

Scheduling a daily backup

Step	Action
1	Click the Schedule Backup tab of the TPN Backup / Restore page. Schedule Backup tab appears.
2	Select a backup configuration ID from the Available Configurations list. The selected backup configuration ID is highlighted. This is the backup configuration for which you will be creating a daily backup.
3	Click the Schedule button. Schedule Backup Wizard appears.
4	Type in the task name of the scheduled backup. The name that you choose should reflect the nature of the scheduled backup. Some examples of possible names are: Midnight Daily , 10PM Daily , or 4AM Daily .
5	Click Daily .
	ATTENTION To run the backups under a specific user account, enter the account name in the Run As User text box, and type the password in both the Password and Confirm password boxes. The account must be a member of either the Administrators or TPS Engineers group. Also, the passwords are only validated for consistency; this dialog does not verify that the named account and passwords authorize the running of the task.

4. Using the TPN Backup and Restore Tool

4.3 Scheduling Backups

Step	Action
	If the Run As User text box is left blank, the Windows Task Scheduler will attempt to run the backup under the default (local SYSTEM) account, but the standard security policies will block the execution.
6	Type in user name in Run As User text box.
7	Type in password and re-enter password.
8	Click Next button. The next page of the backup wizard appears.
9	Select the Start time using the up/down buttons. Selected Start time appears in box.
10	Select the interval (Every Day , Weekdays , or Every x days) at which this backup is to be performed.
11	Select the Start date combo box. Calendar appears on screen.
12	Select the date when the first backup is to be performed. The combo box is updated with the selected date.
13	Click Next button. Schedule details appear.
14	Review the schedule details. If you have to change anything in the schedule, use the Back button.
15	Select Finish button . You have now completed the backup schedule for a daily backup.

Using the Cancel Button

Selecting the **Cancel** button will cancel the entries made on the current screen.

4. Using the TPN Backup and Restore Tool

4.3. Scheduling Backups

Scheduling a Weekly Backup

Step	Action
1	Click the Schedule Backup tab of the TPN Backup / Restore page. Schedule tab appears.
2	Select a backup configuration ID from the Available Configurations list. The selected backup configuration ID is highlighted. This is the backup configuration for which you will be creating a weekly backup.
3	Select the Schedule button. The Schedule Backup Wizard appears.
4	Type in the name of the scheduled backup. The name that you choose should reflect the nature of the scheduled backup. Some examples of possible names are: Weekly Backup or 4AM Sunday and Thursday Backups .
5	Select Weekly .
6	Type in user name in Run As User text box.
7	Type in password and re-enter password.
8	Select Next button. The next page of the wizard appears.
9	Select the Start time using the up/down buttons. Selected Start time appears in box.
10	Select the weekly interval (1, 2, 3...) for Every X Weeks between each backup using the up/down buttons.
11	Select the day(s) of the week when the backup is to occur.
12	Select Next button.
13	Review the schedule details. If you have to change anything in the schedule, use the Back button to go to where you want to make the changes.
14	Select Finish button. You have now completed the backup schedule for a weekly backup.

Scheduling a Monthly Backup

Step	Action
1	Select the Schedule Backup tab of the TPN Backup / Restore page so that the tab contents appear.
2	Select a backup configuration ID from the Available Configurations list. The selected backup configuration ID is highlighted. This is the backup configuration for which you will be creating a daily backup.
3	Select the Schedule button. The Schedule Backup Wizard appears.
4	Type in the task name of the scheduled backup. The name that you choose should reflect the nature of the scheduled backup. Some examples of possible names are: Monthly Backup , 6 Month Backup , or 2 Month Backup .
5	Select Monthly .
6	Type in user name in Run As User text box.
7	Type in password and re-enter password.
8	Select Next button. The next page of the wizard appears.
9	Select the Start time using the up/down buttons. Selected Start time appears in box.
10	Select Day and then select the numerical day of the month such as 1-30 using the up/down buttons Or Select the start time and select the day in the week using the drop-down lists. A sample entry would be: "first Monday," "last Friday," or "second Saturday."

4. Using the TPN Backup and Restore Tool

4.3. Scheduling Backups

Step	Action
11	Select the month(s) when the backup is to occur.
12	Select Next button. The details of the backup schedule appear.
13	Review the backup schedule details. If you have to change anything in the schedule, use the Back button to go to where you want to make the changes.
14	Select Finish button. You have now completed the backup schedule for a monthly backup.

Modifying the Schedule of a Backup

Step	Action
1	Select the Schedule Backup tab of the TPN Backup / Restore page. The Schedule Backup tab appears.
2	Select the schedule you want to modify from the Scheduled Configurations list. The selected backup configuration ID is highlighted.
3	Click the Schedule button. The Schedule Backup Wizard appears.
4	Modify the schedule in accordance with the previous procedures.
5	You can also change the user credentials at this time
6	Review the revised backup schedule. If you have to change anything in the schedule, click the Back button to go to where you want to make the changes.
7	Click Finish button.

Unscheduled a backup



ATTENTION

When you unschedule a backup that is currently executing, the backup in progress continues until it is completed but it is no longer scheduled.

Step	Action
1	Select the Schedule Backup tab of the TPN Backup / Restore page. Schedule Backup tab appears.
2	Select the schedule you want to delete from the Scheduled Configurations list. The selected schedule name is highlighted.
3	Click the Unschedule button. A pop-up message appears stating that: The command cannot be undone. Are you sure you want to Delete "Filename"?
4	Click Yes . Selected schedule is deleted from the Scheduled Configurations list.

4.4 Demanding a backup

Step	Action
1	Select the Schedule Backup tab of the TPN Backup / Restore page. Schedule backup tab appears.
2	Select the configuration ID of the backup from the Available Configurations list. The selected configuration ID name is highlighted.

4. Using the TPN Backup and Restore Tool

4.5. Monitoring backup progress

3	Click the Demand button. Selected backup is initiated, and the backup-in-progress icon appears on the system tray during the entire backup operation to indicate that a backup is in progress. Refer to the <i>TPN Backup/Restore Icon</i> section for a description of the icon.
----------	--

4.5 Monitoring backup progress

Step	Action
1	Right-click on the backup-in-progress icon in the system tray. (Refer to the <i>TPN Backup/Restore Icon</i> section for a description of the icon.) Menu appears.
2	Click Backup Status on the menu.
3	Status/Progress dialog appears. The dialog shows the file currently being processed and the list of detected errors. You can also stop the backup by clicking Cancel backup.

4.6 Deleting a backup

The Available Backups list provides you with a list of available backup IDs from which you can select a backup to delete. If no backups have been performed, the list is blank.

The same backup ID can appear any number of times in the list but each appearance has a different time stamp. A yellow exclamation-mark icon only appears in front of the backup IDs that had errors detected during the backup process.

4. Using the TPN Backup and Restore Tool

4.6 Deleting a backup

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
2	Select the name of the backup that you want to delete from the backups in the Available Backups list. Selected backup is highlighted.
3	Click the Delete Backup button A pop-up message appears stating that: The command cannot be undone. Are you sure you want to Delete (backup ID)?
4	Click Yes to: • Delete the backup file from the Available Backups list. • Delete volumes list in the Scope of Backup window. • Delete the log file. Click Cancel to cancel the delete backup command.
5	If a .LCN file exists for the backup, an additional message appears. Click Yes to delete the .LCN file. Click No to delete everything except the .LCN file.

4. Using the TPN Backup and Restore Tool

4.7. Creating .LCN Files

4.7 Creating .LCN Files

Creating a .LCN file from a backup file

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
2	Select a backup from the Available Backups list.
3	Select the scope of the .LCN file by selecting one or more backup volumes/directories from the Scope of Backup list. or Select the Select Entire Scope button to select all backup files in the list. Each selected backup volume/directory/file is highlighted.
4	Click the Create .LCN File button. Create .LCN File dialog box appears.
5	Type in a full pathname for the new .LCN file such as: C:\LCNS\Nopersonalities.lcn or use the Browse button to find a pathname. Pathname appears in dialog box.
6	Click the OK button. .LCN files are created from the selected backup volumes/directories and are written (in directory form) to the location specified by the pathname.

Viewing information about created .LCN files

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Available .LCN Files tab. The Available .LCN Files tab appears.

4. Using the TPN Backup and Restore Tool

4.8 Viewing Log Files created during backup or .LCN File creation

2	Select a .LCN file from the .LCN Files Available list. Selected .LCN file is highlighted. The scope of the .LCN file is shown in Scope Of .LCN File list. The full pathname of the .LCN file is shown in the .LCN File Name textbox. The related backup source for the .LCN file is shown in the Backup Source For .LCN File textbox.
3	To see the list of directories, click on the + next to the !!!! volume name to expand.
4	Repeat step 3 to expand the directories into files until you find the file that you want. Note that files at the volume level appear in separate directories. All file have the volume designator !!!!.

4.8 Viewing Log Files created during backup or .LCN File creation

Step	Action
1	Click on Backup Image Operations tab or Available .LCN Files tab.
2	Select a backup from the Available Backups list (Backup Image Operations tab) or select a .LCN file from the Available .LCN Files list (Available .LCN Files tab).
3	Click on View Log File button. Log files appear as shown below, depending on the logging level configured for the backup.

4. Using the TPN Backup and Restore Tool

4.8. Viewing Log Files created during backup or .LCN File creation

Backup Log File configured for Volume-level

The volume-level log file contains the following items as shown in Figure 3-3

- Log start time
- Log end time
- Location of backup files
- Location of .LCN file (if requested)
- Optional checkpoint pass/fail status for each checkpointed node, volumes included
- Errors

Backup Path: C:\HWIAC\Backup\DataVols 01-May-2004 010000

LCN File Path: C:\LCNS\DataVols.LCN

01May2004 01:00:00 INFO Backup Started

01May2004 01:00:00 OK Checkpoint of Node 23 Completed

01May2004 01:00:00 ERROR Checkpoint of Node 26 Failed

01May2004 01:00:23 INFO Volume ABCD Started

01May2004 01:15:52 OK Volume ABCD Completed 0000 Errors 00056 Files Copied

01May2004 01:15:56 ERROR Backup Completed 0001 Errors 00056 Files Copied

01May2004 01:15:59 OK LCN File Creation Started

01May2004 01:30:55 OK LCN File Creation Completed 0000 Errors 00056 Files Copied

Figure 4-1 Volume-level log file example

Backup Log File configured for Directory-level

The directory-level log file contains the following items as shown in Figure 3-4

- Log start time
- Log end time
- Location of backup files
- Location of .LCN file (if requested)
- Optional checkpoint pass/fail status for each checkpointed node
- Volumes included
- Directories included
- Errors

```
Backup Path: C:\HWIAC\Backup\DataVols 01-May-2004 010000
LCN File Path: C:\LCNS\DataVols.LCN
01May2004 01:00:00 INFO  Backup Started
01May2004 01:00:00 OK    Checkpoint of Node 23 Completed
01May2004 01:00:00 OK    Checkpoint of Node 26 Completed
01May2004 01:00:23 INFO  Volume ABCD Started
01May2004 01:00:24 INFO  Directory EFGH Started
01May2004 01:15:30 OK    Directory EFGH Completed  0000 Errors  00056 Files Copied
01May2004 01:15:52 OK    Volume ABCD Completed    0000 Errors  00056 Files Copied
01May2004 01:15:56 OK    Backup Completed          0000 Errors 00056 Files Copied
01May2004 01:15:59 OK    LCN File Creation Started
01May2004 01:30:55 OK    LCN File Creation Completed 0000 Errors  00056 Files Copied
```

Figure 4-2 Directory-level log file example

4. Using the TPN Backup and Restore Tool

4.8. Viewing Log Files created during backup or .LCN File creation

Backup Log File configured for File-level

The file-level log file contains the following items as shown in Figure 3-5

- Log start time
- Log end time
- Location of backup files
- Location of .LCN file (if requested)
- Optional checkpoint pass/fail status for each checkpointed node
- Volumes included
- Directories included
- Files included
- Errors

Backup Path: C:\HWIAC\Backup\DataVols 01-May-2004 010000

LCN File Path: C:\LCNS\DataVols.LCN

01May2004 01:00:00 INFO Backup Started

01May2004 01:00:00 OK Checkpoint of Node 23 Completed

01May2004 01:00:00 OK Checkpoint of Node 26 Failed

01May2004 01:00:23 INFO Volume ABCD Started

01May2004 01:00:23 INFO Directory EFGH Started

01May2004 01:00:23 INFO File ABCDEFGH.IJ Completed

01May2004 01:15:48 OK Directory EFGH Completed 0000 Errors 00056 Files Copied

01May2004 01:15:52 OK Volume ABCD Completed 0000 Errors 00056 Files Copied

01May2004 01:15:56 OK Backup Completed 0000 Errors 00056 Files Copied

01May2004 01:15:59 OK LCN File Creation Started

01May2004 01:30:55 OK LCN File Creation Completed 0000 Errors 00056 Files Copied

Figure 4-3 File-level log file example

.LCN Log File

The .LCN log file contains the following items as shown in Figure 3-6

- Log start time
- Log end time
- Location of backup files
- Location of .LCN file
- Errors

Backup Path: C:\HWIAC\Backup\DataVols 01-May-2004 010000

LCN File Path: C:\LCNS\DataVols.LCN

01May2004 01:15:59 OK LCN File Creation Started

01May2004 01:30:55 OK LCN File Creation Completed 0000 Errors 00056 Files Copied

Figure 4-4 .LCN log file example

4. Using the TPN Backup and Restore Tool

4.8. Viewing Log Files created during backup or .LCN File creation

5. Restoring Volumes, Directories, and Files

5.1 About this section



ATTENTION

The TPN Backup and Restore tool will not restore !9xx volumes or History Group Definition Files (APL*.MM) on the !0xx volumes, or any files in the &ASY directory that already exist.



ATTENTION

The TPN Backup and Restore tool will not restore file descriptors because File Transfer does not copy file descriptors.

5.2 Recovering the History Module

If the HM disk drive has been replaced, you must get the HM in a state where the TPN Backup and Restore function's restore capabilities can restore the remaining HM files. You must first perform a procedure similar to: "Restoring the HM Group Definition Files (APL) without the Continuous History Data" procedure as defined in the *Engineer's Reference Manual*. This procedure is provided below.

Step	Action
1	Replace the failed HM drive in accordance with the procedures in the <i>History Module Service</i> manual.
2	Start the HM in the INIT (HMOF) personality. HMOF appears as the HM status on the Node Status Display.
3	Initialize the HM. Initializing the HM creates the Lxx_NVCF and Cxx_CPNT files on the !9xx volume.

5. Restoring Volumes, Directories, and Files

5.2. Recovering the History Module

Step	Action
4	Restore the APL*.MM files using the RSHSDFZ.EC file if the HM contains continuous history.
5	Run the LOC_VOLZ.EC file to make the HM bootable which involves writing a boot record and on-line personality files to the !9xx volume.
6	Restore the NCF files in the &ASY volume if the HM is the system HM.
7	Autoboot the HM to place it in the on-line (HMON) personality. HM is now ready to have the volumes restored.
8	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
9	Select a backup ID from the list of Available Backups . List of volumes in selected backup appears in the Scope of Backup window.
10	Select Deselect Entire Scope button. All volumes in list are deselected.
11	Select all the volumes that are to reside on the HM drive.
12	Select the Restore Backup button. Restore Backup dialog box appears.
13	Select the log file mode (Volume , Directory , or File) from the Log File Mode drop-down list box.
14	If you do not want to overwrite existing or protected files during the restore operation, select the Do not overwrite existing or protected files box . If you select this box, go to step 17. If you deselect this box perform steps 15 and 16. If the box is deselected, you must make additional selections in the Overwrite Existing / Protected Files options portion of the dialog box to determine the actions performed on the existing files and protected files during the restore.

5. Restoring Volumes, Directories, and Files

5.3 Restoring selected volumes

Step	Action
15	For the Existing Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
16	For the Protected Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
17	Select the Restore button. The restore volume operation is started and the Progress Prompt appears. The Progress Prompt disappears when the restore operation is finished.

5.3 Restoring selected volumes

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
2	Select a backup ID from the list of Available Backups . List of volumes in selected backup appears in the Scope of Backup window.
3	Select Deselect Entire Scope button. All volumes in list are deselected.
4	Select only the volumes that are to be restored from the Scope of Backup list.
5	Select the Restore Backup button. Restore Backup dialog box appears.
6	Select Volume, Directory, or File from the Log File Mode drop-down list box.

5. Restoring Volumes, Directories, and Files

5.4. Restoring selected directories

Step	Action
7	If you do not want to overwrite existing or protected files during the restore operation, select the Do not overwrite existing or protected files box. If you select this box, go to step 10. If you deselect this box perform steps 8 and 9. If the box is deselected, you must make additional selections in the Overwrite Existing / Protected Files options portion of the dialog box to determine the actions performed on the existing files and protected files during the restore.
8	For the Existing Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always.
9	For the Protected Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always.
10	Select the Restore button. The restore volume operation is started and the Progress Prompt appears. The Progress Prompt disappears when the restore operation is finished.

5.4 Restoring selected directories

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
2	Select a backup ID from the list of Available Backups. List of volumes in selected backup appears in the Scope of Backup window.
3	Select Deselect Entire Scope button. All volumes in list are deselected.

5. Restoring Volumes, Directories, and Files

5.4 Restoring selected directories

4	Select only the volumes that are to be restored from the Scope of Backup list.
5	Select the Restore Backup button. Restore Backup dialog box appears.
6	Select Volume from the Log File Mode drop-down list box.
7	If you do not want to overwrite existing or protected files during the restore operation, select the Do not overwrite existing or protected files box . If you select this box, go to step 10. If you deselect this box perform steps 8 and 9. If the box is deselected, you must make additional selections in the Overwrite Existing / Protected Files options portion of the dialog box to determine the actions performed on the existing files and protected files during the restore.
8	For the Existing Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
9	For the Protected Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
10	Select the Restore button. The restore directory operation is started and the Progress Prompt appears. The Progress Prompt disappears when the restore operation is finished.

5. Restoring Volumes, Directories, and Files

5.5. Restoring selected files

5.5 Restoring selected files

Step	Action
1	With the TPN Backup / Restore page on the screen, select the Backup Image Operations tab. Backup Image Operations tab appears.
2	Select a backup ID from the list of Available Backups . List of volumes in selected backup appears in the Scope of Backup window.
3	Select Deselect Entire Scope button. All volumes in list are deselected.
4	Select the + sign of the volume in the Scope of Backup list that contains the files that you want to restore.
5	Select the + sign of the directory in the Scope of Backup list that contains the files that you want to restore.
6	Select the files from Scope of Backup list.
7	Select the Restore Backup button. Restore Backup dialog box appears.
8	Select File from the Log File Mode drop-down list box. The file restore events will be logged.
9	The existing and protected files will be overwritten during the restore operation unless you select the Do not overwrite existing or protected files box . If you select this box, go to step 12. If you deselect this box perform steps 10 and 11. If the box is deselected, you must make additional selections in the Overwrite Existing / Protected Files options portion of the dialog box to determine the actions performed on the existing files and protected files during the restore.

5. Restoring Volumes, Directories, and Files

5.5 Restoring selected files

10	For the Existing Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
11	For the Protected Files section, select the Overwrite Always or Prompt to Overwrite or Skip Always radio button. The default is Overwrite Always .
12	Select the Restore button. The restore file operation is started and the Progress Prompt appears. The Progress Prompt disappears when the restore operation is finished.

5. Restoring Volumes, Directories, and Files

5.5. Restoring selected files

6. Error Handling

6.1 About this section

This section lists and describes the errors that can occur during backup and during restore operations, and during .LCN file creation

6.2 Errors during backup

Errors that occur during the backup process are listed in Table 6-1 and are included in the log file associated with the backup. If the backup process includes the creation of a .LCN file, errors that occur during the .LCN file creation process will also be included in the log file associated with the backup.

6. Error Handling

6.2. Errors during backup

Table 6-1 Backup Error Messages

Error Message	Description
01May2004 01:00:00 ERROR Checkpoint of Node 26 Failed	Checkpoint of data owner failed.
01May2004 01:00:00 ERROR Volume ABCD Not Found On NET	Volume not found on History Module.
01May2004 01:00:00 ERROR File Transfer Not Active	File Transfer not active.
01May2004 01:00:00 ERROR Volume ABCD Not Configured For File Transfer	Volume not configured for File Transfer.
01May2004 01:00:00 ERROR File ABCDEFGH.IJ Not Configured For File Transfer	File type not configured for File Transfer.
01May2004 01:00:00 ERROR File ABCDEFGH.IJ Opened Exclusively	File opened exclusively.
01May2004 01:00:00 ERROR Backup Not Completed	PC failed during backup.
01May2004 01:00:00 ERROR Backup Aborted By User	Backup aborted by user.
01May2004 01:00:00 ERROR Insufficient disk space	Destination for the backup files has become full.
01May2004 01:00:00 ERROR LCN File Exceeds 500MB	Maximum size of 500MB exceeded during .LCN file creation.
01May2004 01:00:00 ERROR Volume ABCD Not Found In Backup	Volume not found in backup (during .LCN file creation).
01May2004 01:00:00 ERROR LCN File Creation Not Completed	PC failed during .LCN file creation.

6.3 Errors during .LCN file creation

Errors that occur during the .LCN file creation process will be included in the log file associated with the creation of the .LCN file. These error conditions include but are not limited to the errors listed in Table 6-2.

Table 6-2 Errors During .LCN File Creation

Error Message During .LCN File Creation	Definition
01May2004 01:00:00 ERROR LCN File Exceeds 500MB	Maximum size of 500MB exceeded.
01May2004 01:00:00 ERROR Volume ABCD Not Found In Backup	Volume not found in backup.
01May2004 01:00:00 ERROR LCN File Creation Not Completed	PC failed during .LCN file creation.
01May2004 01:30:55 ERROR LCN File In Use – Could Not Be Overwritten	File is mounted somewhere else.

6. Error Handling

6.4. Errors during restore

6.4 Errors during restore

Errors that occur during a restore are included in the restore log file if restore logging has been enabled.

These error conditions include, but are not limited to, the following error condition:

- Checksums failed because the backup files have been modified before the user requested a restore.

Additionally, volumes/files that are skipped during the restore process are also listed in the restore log file, if restore logging has been enabled.

These error conditions include but are not limited to the following conditions:

- !9xx volume was skipped since it already exists.
- File APLxxxxx.MM in the !0xx volume was skipped since it already exists.
- File xxxxxxxx.xx in the &ASY directory was skipped since it already exists.

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